LAB EXPERIMENT RECORD

Set 28: Vehicle Performance Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
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AIM

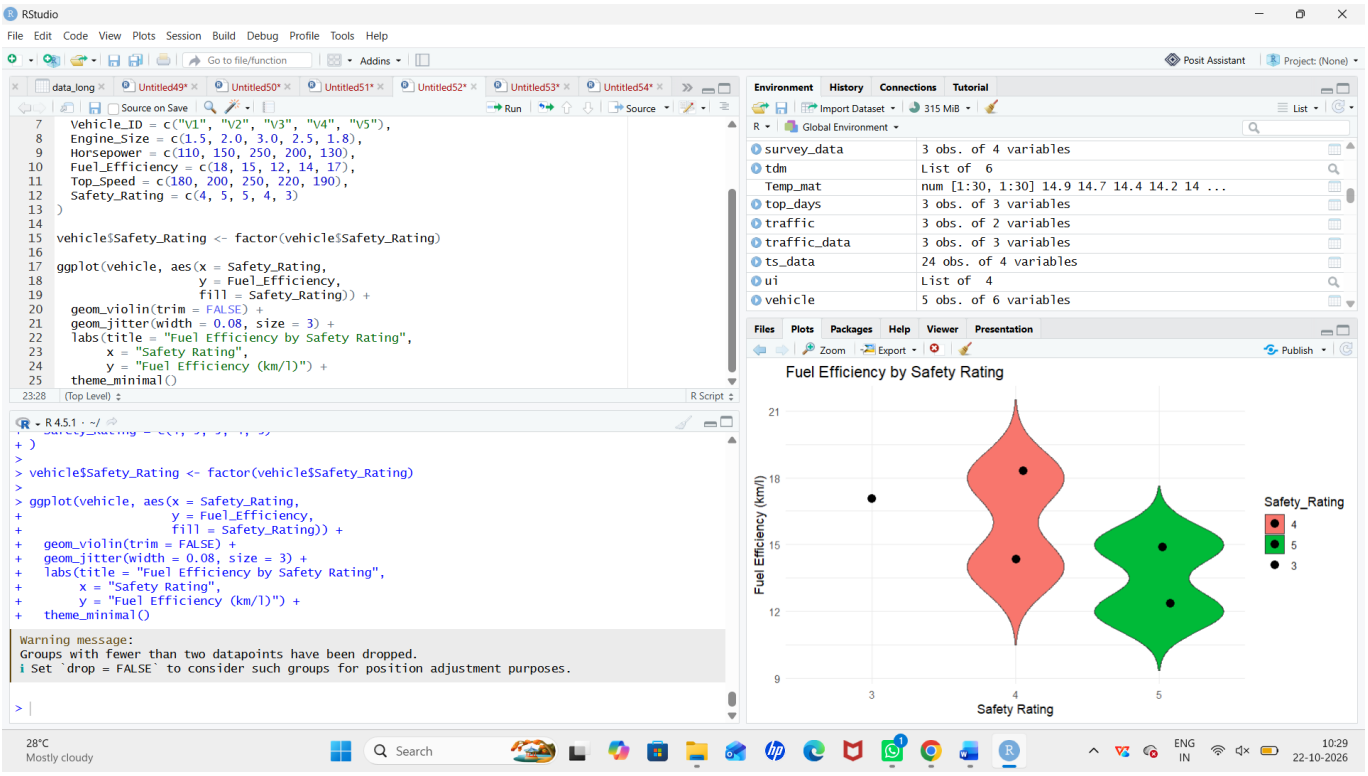
To analyze vehicle performance using R and Tableau by creating a violin plot, scatter plot, correlation heatmap, and interactive dashboard to understand the relationship between fuel efficiency, horsepower, engine size, top speed, and safety rating.

1.

```
# Install package once if required
# install.packages("ggplot2")

library(ggplot2)

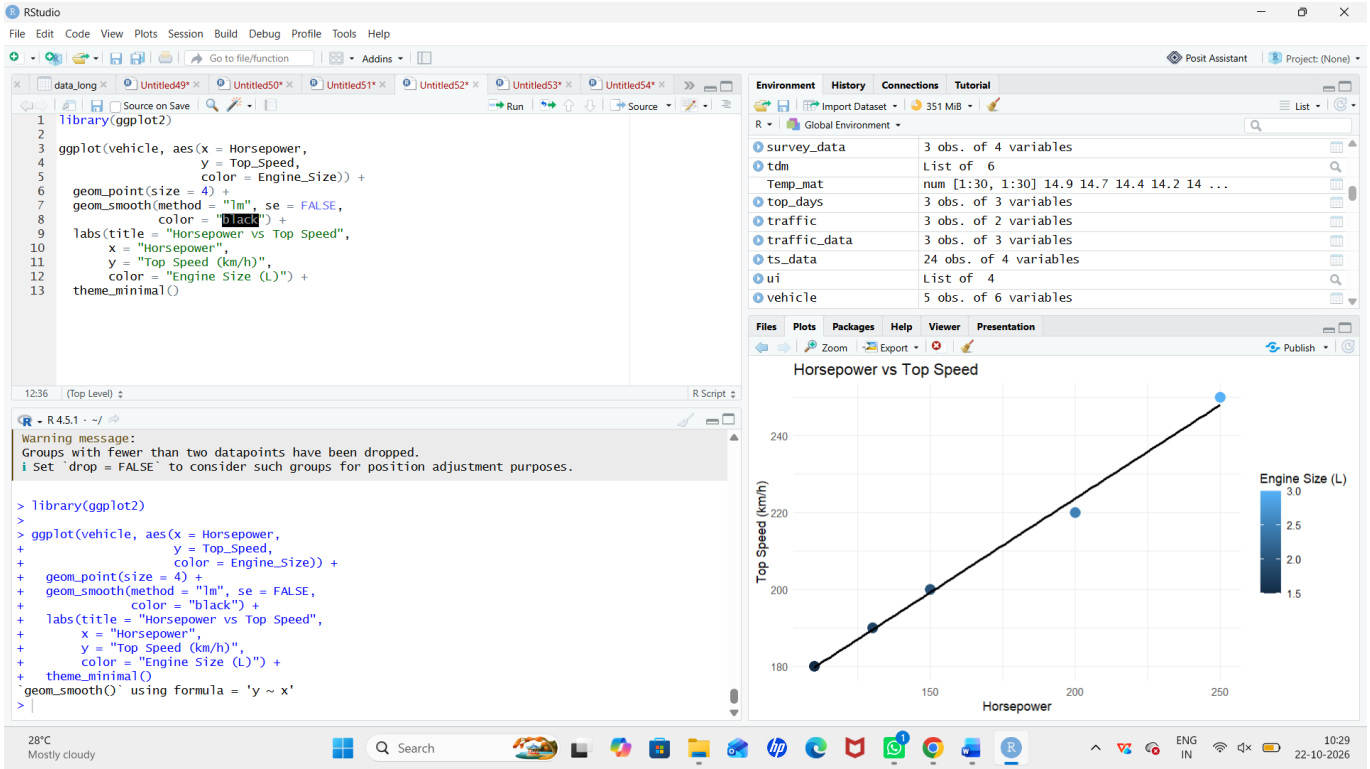
vehicle <- data.frame(
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)
```



2.

```
library(ggplot2)

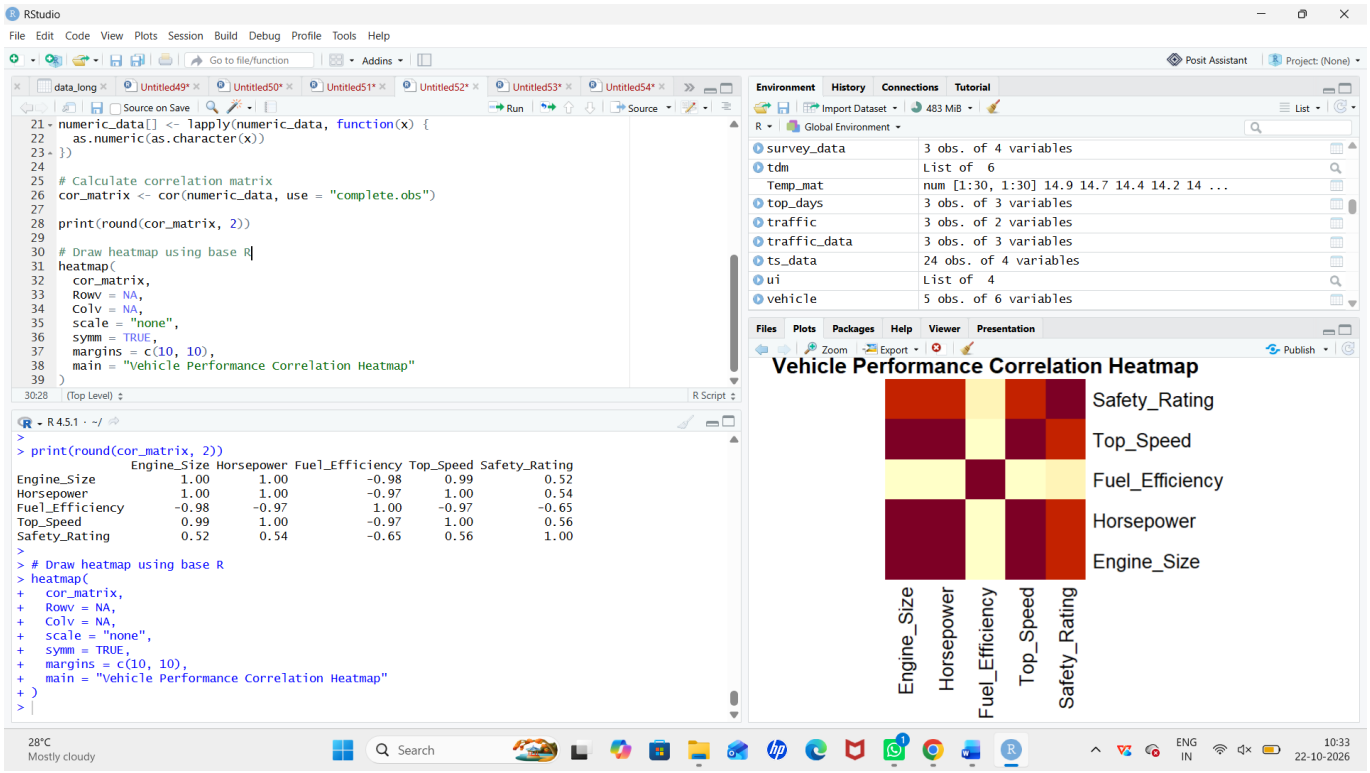
ggplot(vehicle, aes(x = Horsepower,
                    y = Top_Speed,
                    color = Engine_Size)) +
  geom_point(size = 4) +
  geom_smooth(method = "lm", se = FALSE,
              color = "black") +
  labs(title = "Horsepower vs Top Speed",
       x = "Horsepower",
       y = "Top Speed (km/h)",
       color = "Engine Size (L)") +
  theme_minimal()
```



3.

```
# Create the vehicle dataset
vehicle <- data.frame(
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3)
)

# Select numerical variables
numeric_data <- vehicle[, c(
  "Engine_Size",
  "Horsepower",
  "Fuel_Efficiency",
  "Top_Speed",
  "Safety_Rating"
)]
```



4.

Steps to create the dashboard

1. Open Tableau Desktop or Tableau Public.
2. Create a CSV file containing the five vehicle records and connect to it.
3. Open Sheet 1 and create a bar chart:
    - Drag Vehicle_ID to Columns.
    - Drag Fuel_Efficiency to Rows.
    - Choose Bar from the Marks dropdown.
    - Drag Safety_Rating to Color.
4. Open Sheet 2 and create a scatter plot:
    - Drag Horsepower to Columns.
    - Drag Top_Speed to Rows.
    - Drag Vehicle_ID to Detail.
    - Drag Engine_Size to Color.
    - Drag Fuel_Efficiency to Size.
5. Add filters:
    - Drag Engine_Size to the Filters shelf, select the required values, and choose Show Filter.
    - Drag Safety_Rating to the Filters shelf and choose Show Filter.

6. Create a new dashboard and drag both worksheets onto it.

7. Set a suitable title: Vehicle Performance Analysis Dashboard.

8. Test the filters to compare vehicles with different engine sizes and safety ratings.

Dashboard interpretation:

- V1 offers the highest fuel efficiency at 18 km/l among the vehicles with safety rating 4.

- V2 and V3 have safety rating 5, with V3 providing the highest top speed but the lowest fuel efficiency.

- V4 provides a top speed of 220 km/h with fuel efficiency of 14 km/l.

- V5 has fuel efficiency of 17 km/l and safety rating 3, so its lower safety rating should be considered when comparing vehicles.

The dashboard helps users compare fuel efficiency, speed, horsepower, and safety rating to identify vehicles that balance economy and performance. Safety ratings and efficiency figures should be considered together rather than selecting a vehicle based on speed alone.

PROCEDURE

1. Create the vehicle dataset in R with engine size, horsepower, fuel efficiency, top speed, and safety rating.

2. Use ggplot2 to generate a violin plot and scatter plot.

3. Calculate correlations among the numerical variables and display them using a heatmap.

4. Import the dataset into Tableau and build bar charts and scatter plots.

5. Add engine-size and safety-rating filters and combine the charts into an interactive dashboard.

6. Interpret the results to compare vehicle efficiency and performance.

RESULT

Vehicle performance was analyzed using R and Tableau. The plots showed a positive relationship between horsepower and top speed, while higher horsepower was associated with lower fuel efficiency in this small dataset. The correlation heatmap summarized relationships among the numerical variables, and the Tableau dashboard enabled interactive comparisons to identify vehicles with different combinations of fuel efficiency, speed, and safety rating.